

Qiyuan (Vincent) Zhu

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EDUCATION

Northeastern University

Master of Computer Science

Sep 2023-May 2026

University of Pennsylvania

Master of Public Administration

Sep 2022-May 2023

Beijing Foreign Studies University

Bachelor of Arts in Portuguese Studies

Sep 2017-Jun 2021

TECHNICAL SKILLS

Programming Languages: Java, JavaScript/TypeScript, Python

Frameworks/Tools: Backend - Springboot/Node.js/FastAPI, ORM – Hibernate/Mybatis; Micro Services; API-SOAP/REST/RPC, Converter - Jackson2; Dependency – Maven/Gradle; Git – GitHub, Bitbucket; Docker; Linux - Command, CronJob; CI/CD - AWS CodeCommit, CodeBuild, ECR, CodeDeploy; Message Queue - Kafka, AWS

Kinesis/SNS/SQS; Relational Database - MySQL, PostgreSQL, AWS RDS; NoSQL - MongoDB, AWS DynamoDB;

Serverless - AWS Lambda; Cache - Redis; Unit Test - Junit; Monitoring & Dashboard - Splunk

Cloud Platform: AWS(S3, EC2, RDS, DynamoDB, Lambda, Kinesis, SNS, SQS, RedShift, Glue, IAM, EMR, CodeCommit, CodeBuild, ECR, CodeDeploy etc.), Azure, GCP

PROFESSIONAL EXPERIENCES

Shawk Solution

Software Engineer Intern

Oct 2025-Dec 2025

- Built a full-stack education platform using **React**, **TypeScript**, and **Tailwind CSS**, delivering a responsive UI that improved page load speed by **30%** across devices.
- Implemented secure authentication with **JWT** and **bcrypt**, integrating **WeChat QR login**, which increased user sign-up rate by **20%**.
- Designed and deployed a **PostgreSQL** database schema, optimizing queries via **Spring Data JPA** and **Hibernate ORM**, which reduced average query latency by **25%**.
- Developed Java Spring Boot **RESTful APIs** with **90%** unit test coverage, integrated **GitHub Actions CI/CD** and **AWS Cloud Watch** monitoring, cutting regression bugs by **30%** and improving deployment efficiency.

Artisk Inc.

Software Engineer Intern

Jul 2025-Sep 2025

- Architected a full-stack payment system using **Stripe Checkout API** and **Meteor**, supporting both one-time purchases and recurring subscriptions with embedded checkout flow, processing 3 product tiers for logo design credits.
- Engineered Stripe **webhook pipeline** handling **6+ event types** with signature verification, idempotent processing via **MongoDB** deduplication, and retry logic (exponential backoff, max 4 attempts), achieving **99.9%** payment fulfillment reliability.
- Designed and implemented **6 MongoDB collections** with **SimpleSchema validation** and audit trail logging, enabling real-time subscription lifecycle management and automated design-attempt allocation.

PROJECT HIGHLIGHTS

Video RAG Agent System

Dec 2025-Feb2026

- Built a multi-user **Video RAG SaaS** platform with **Node.js/Express**, **PostgreSQL**, and **Docker Compose**, implementing **JWT** authentication, **bcrypt** hashing, role-based data isolation, and 3-tier rate limiting, supporting concurrent processing of **3** videos per user with sub-second semantic search across **500+** text chunks.
- Designed an **asynchronous** media processing backend handling **YouTube URL** ingestion and file uploads up to 100MB, integrating **FFmpeg** audio extraction, **Whisper** transcription, and **OCR** frame analysis with real-time progress streaming, reducing end-to-end video-to-knowledge pipeline from manual hours to under **5 minutes** per 10-minute video.
- Deployed on **AWS EC2** with **Docker Compose** and **Cloudflare CDN/SSL**, optimized memory footprint by **70%** using CPU-only PyTorch (900MB → 200MB) and Whisper tiny model with **2GB** swap, achieving production uptime on **1GB RAM** at **<\$6/month** hosting cost.

Distributed Task Scheduling Platform

Aug 2025-Oct 2025

- Built a **distributed task scheduling platform** with **3 Spring Boot microservices** (Scheduler, Worker, Common), using **Redis-based** leader election and distributed locks to ensure high-availability scheduling across multiple nodes.
- Designed a flexible **execution engine** supporting Standalone, Broadcast, and Sharding modes with configurable **retry**, **timeout**, and **Cron scheduling**, dispatching tasks via Redis queues to a horizontally scalable worker pool.
- Developed a **React + TypeScript** management dashboard with Ant Design Pro, featuring real-time cluster monitoring, task **CRUD** operations, execution history tracking, and 24-hour success/failure trend visualization.

Product Information Storage and Publisher System

Jun 2025-Aug 2025

- Built a new product publishing system connecting **Vue.js** with **MySQL**, **AWS SNS/SQS**, and **3 Spring Boot consumers**, automating data flow across systems.
- Developed an end-to-end **data pipeline** with scheduled **Cron jobs** and **fan-out messaging**, enabling daily synchronization to multiple consumers without manual intervention.
- Deployed **Spring Boot** apps on **AWS EC2**, configured **Spring AOP** logging, and integrated **Splunk Dashboards** with **AWS CloudWatch** for real-time monitoring.